



Dug Out: Archaeologists complete excavation of former Leetsdale island

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By Maria Carpico, Post-Gazette Staff Writer

fter nearly three years of digging, the excavation is finished at a Leetsdale site that is revealing how people labored 8,000 years ago.

Now the analysis of the uncovered artifacts and data begins.

The site, a section of Leetsdale Industrial Park that once was a sandbar in the Ohio River, contains "the deepest stratified dig of its kind in Pennsylvania," said Frank Vento, a geology professor at Clarion University.

On Monday, archaeologists, geologists, botanists, geophysicists and volunteers wrapped up their excavating work at what turned out to be a treasure trove of preserved history.

That's because the former sandbar, which eventually became an island and then attached itself to the mainland, had never been plowed or dug up, allowing archaeologists to locate perfectly segmented layers. Also, when the site flooded, soil was deposited over the existing layer, effectively sealing it and everything underneath.

"It's a real advantage that [the layers are] separated. We can see where they went from hunters and gatherers to growing their own food," said Patricia Miller, principal investigator for KCI Technologies in Mechanicsburg, which was contracted to complete the work and a comprehensive report on the project.

As recently as May 21, in the closing days of the dig, two stone tools were found, which prompted archaeologists to continue work for another week.



Igor Nurabas, left, of KCI Technologies, uses a trowel to skim thin layers of soil from the bottom of an archaeological dig along the Ohio River in Leetsdale. (Bob Donaldson, Post-Gazette)

"A common artifact is 'chippage' from the production of stone tools," said Kurt Carr, division chief of archaeology and protection for the Pennsylvania Bureau of Historic Preservation. Chippage consists of the chips of stone that result from the making of arrowheads and spears. "It's not a spectacular find, but it tells you something about the inhabitants."

The site was discovered through an archaeological survey that was conducted before the Pittsburgh District of the Army Corps of Engineers began a project to replace the Braddock Dam. The corps had the dam built on the banks of the Ohio River in Leetsdale and then floated into place on the Monongahela River.

To comply with the National Historic Preservation Act, the archaeological survey was required before work on the dam project could begin in 1999.

"We recognized that we were going to impact the site and slightly redesigned our plan to include a fairly extensive public outreach plan," said Conrad Weiser, environmental resources planner for the corps.

Excavations began in fall of 2000 on the first of three sections of the site and yielded the discovery of the Harmony Brick Works, a factory that operated from the late 1800s to the early 1900s. The brick factory was built and run by the Harmony Society, whose community, Old Economy Village in Ambridge, has been preserved by the Pennsylvania Historical and Museum Commission.

While digging in another section of the site, archaeologists located steatite, or soapstone discs that are not characteristic of southwestern Pennsylvania, but are commonly found in southeastern Pennsylvania and Maryland.

"Soapstone was soft and used for making bowls," Miller said. Researchers believe inhabitants from the eastern part of the state migrated to the area.

An 8-pound sandstone bowl that is about 3,000 years old also was found in that area.

In August, work began on the last section to be excavated.

A few meters of industrial fill were removed and excavation began by going down a meter and then horizontally a meter, until the site became roughly 18 feet deep.

Researchers have determined that portions of the site range between the Early Woodland period, roughly 1,100 B.C. to A.D. 200, and



J.T. Marine measures elevation at an archaeological excavation site in Leetsdale.

the Archaic periods, about 6,000 B.C. to 1,100 B.C.

Exact time periods have not been pinpointed yet because testing has just begun on the soil samples extracted from each strata. The archaeologists' initial findings turned up red and oxidized soils showing evidence of shallow fire pits.

Marine, of KCI Technologies, works with an instrument that uses a laser beam to take measurements. Click photo to view larger version. (Bob Donaldson, Post-Gazette)

"It's wonderful to look at all these areas and the different nuances that occurred during the occupations," said Igor Nurabas, crew chief for KCI who has been on the project from the beginning.

Eventually, comparisons will be drawn between the flood plains to determine what the environment was like when the various groups populated the area. "This provides us with a unique picture of the changes in climate from deglaciation to the present," Vento said.

The state museum commission eventually will handle the long-term documentation of all the artifacts found on the site, and the corps hopes to create a booklet detailing the project from inception to completion.

A different archaeology firm was used for each site, partly because of construction constraints, land access and the project's three-year time line.

"We were nervous about that, but it made for a much better project," Carr said. "The archaeologists all had different backgrounds that added to the interpretations." To maintain continuity, two projectwide conferences were held.

The corps is allowed to spend 1 percent of the project's total cost on data recovery, which is about \$7.05 million. According to Weiser, the plan was redesigned to fall within the 1 percent guideline set by the government. A large percentage of the cost is going to the initial unearthing of the site.

Leetsdale Industrial Park, which owns the site, gave the corps permission to completely investigate all aspects of the area. Now that the digging is finished, the excavated area, which varies from 16 to 18 feet deep, will be back filled, compacted and returned to the industrial park by the end of July.

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[Back](#)

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